

Quiz 1 - Computational Physics I

NAME: Silva Castro Alejandro Fabián SCORE: 18/20

Date: Wednesday 26 February 2025 Duration: 45 minutes
Credits: 20 points (5 questions) Type of evaluation: LAB

Provide short and concise answers to the following items:

1. (4 points) Programming Languages

- (a) Explain the difference between low-level and high-level programming languages.
- (b) Provide 1 example of a low-level programming language and 1 of a high-level language.

A low-level programming language is closer to the computer, so they are hard to understand for us humans. Also they need to be compiled into a binary file to be executed. An example is "C".

A high-level programming language is easier to understand for us humans, so they are the most used. We need an interpreter that executes the script line by line. An example is "python".

2. (4 points) Python Functions

- (a) Explain what a Python function is.
- (b) List 4 types of Python functions.

A python function is an object that receives some input and, after doing some computation, returns some values. We have native python functions, functions native to third-party libraries and we can create our own functions. There are also the lambda functions that are used when the computation does not take so many steps.

Nested functions?
Recursive functions?

3. (4 points) Lambda Functions and Vectorisation

- (a) Briefly explain what a lambda function is in Python.
- (b) Briefly explain the concept of vectorisation in Python.

-0.5 A lambda function is an object that let us write a function computation in a concise manner. We just need to declare the variables and the computation to be done. How? 1-line, single expression with implicit return.

All functions in Python are vectorized, this means that we can (and also are encouraged to) use vectors/arrays as inputs. The computation will be done on each element of the vector. ↳ N-d arrays.

An array is the generalization of a vector in Python.

4. (4 points) Linux File Permissions

In a Linux environment, the file permissions for a script named `run_test.sh` are currently set as `-rw-rw-r--`. You need to allow all users to execute the script but want to keep the current read and write permissions unchanged.

(a) Explain the meaning of the current file permissions `-rw-rw-r--` for `run_test.sh`.

(b) Provide the `chmod` command you would use to modify the permissions so that all users can execute the script without altering the read and write permissions.

(c) After applying the `chmod` command, what will the new permissions of the file `run_test.sh` be? Write them in both symbolic and numeric forms.

a) `-` at the beginning means that we have a file. Then, the user and the group of users who have access to the computer can read and write that file. Also, everyone else (the world) can just read the file.

b) `chmod 775 run_test.sh` (convention: $r=4$, $w=2$, $x=1$)

c) `-rwx-rwx-rx`

The user and group can read, write and execute the file.
Everyone else can just read and execute the file.

5. (4 points) Python I/O and Plotting

In an optics experiment, a laser beam is passed through a single-slit aperture, and the resulting diffraction pattern is recorded on a screen. A photodetector scans across the screen, measuring the light intensity at different positions. The recorded data is stored in a text file named `diffraction_data.txt`, which contains two columns: the first column represents the position on the screen (x); the second column represents the measured light intensity (I).

(a) Sketch an algorithm workflow to read the data from the file, and generate and save a plot of light intensity vs. position using Python.

(b) Write the overall syntax of a native I/O function that reads the data from a given file, and returns the extracted x and I values as NumPy arrays.

a) Read the data. This is not a workflow.

↳ I/O function

Extract the numbers into lists (one list for each column)

Convert the lists into arrays. Plotting I/O function.

Begin the figure environment and plot intensity vs. position (position on x-axis and intensity on y-axis)

Save plot ("filename"):

b) `def io_function("filename"):`
`file1 = open("diffraction_data.txt", "r")`

`data = file1.read()` # this should be `file1.readlines()`
`import numpy as np` # import the library outside of the function

`x = []`

`i = []`

for numbers in `len(data)`
`numbers = data.split()`

`x.append(numbers[0])`

`i.append(numbers[1])`

(continuation)

`x_a = np.array(x)`

`i_a = np.array(i)`

return `x_a, i_a`